UX CASE STUDY · AR MICRO-EXPERIENCE

Pip's Thought Orbit

An AR companion for parking unfinished intentions, and finding the thread again.

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UX DESIGN DOCUMENTATION · 24 OCTOBER 2026



Pausing a task loses the context

Design students pause tasks all the time. A short to-do keeps the title, but not what they were doing or what to do next, so restarting means reconstructing it.

HOW MIGHT WE

help design students park an unfinished intention with enough context to resume it later, using a calm spatial interaction that does not add more cognitive clutter?

TARGET USER

Design students, about 18–25, in Pune

CONTEXT

Studio projects, research and classes; some also intern or work part-time

SETTING

A personal workspace: a desk at home, the studio or the library

NEED

Leave a task with enough context to return to it with less reconstruction

Two small studies, three sources

PRIMARY · INTERVIEW

1 design student, Pune · 19 min

Lists, alarms and her mum's check-ins; still forgets mid-priority tasks. Wants "someone who is actually checking in".

PRIMARY · CUE EXPERIMENT

4 participants

Task headings outlast details; colour becomes a retrieval anchor. "Visuals were the things which made me recall."

GILBERT ET AL. · 2023

People offload more under memory load and interruptions, and it works.

ALTMANN & TRAFTON · 2004

After an interruption, resuming took about 3.8 s vs 1.9 s; earlier cues cut the lag.

WANG & PÉREZ-QUIÑONES · 2014

Reminders aid retrieval, but automatic capture leaves "very fuzzy memories".

Primary data is small-n and directional, not representative.

What the research changed

RESEARCH SAYS		PIP DOES
Reminders cost effort, so busy people skip them Gilbert et al., 2023	→	Parking needs only a title; cues are optional
Headings outlast details; clutter hurts encoding Cue experiment, n = 4	→	One titled orb per thought, three orbs a page
Interruptions add resumption lag; cues cut it Altmann & Trafton, 2004	→	The card restores “what I was doing” and the smallest next step
An alarm isn’t enough; she wants a check-in Interview, n = 1	→	Optional bring-back time on each thought, while Pip is open

From interruption to resuming

STAGE 1

Pause

Switch tasks mid-way

> “I need to switch, but I do not want to lose my place.”

STAGE 2

Park

Title first, cues if time

> “I can record this before I move on.”

STAGE 3

Return

Bring Pip to a surface

STAGE 4

Scan

Orbs ring Pip, three a page

> “My unfinished thoughts are visible here.”

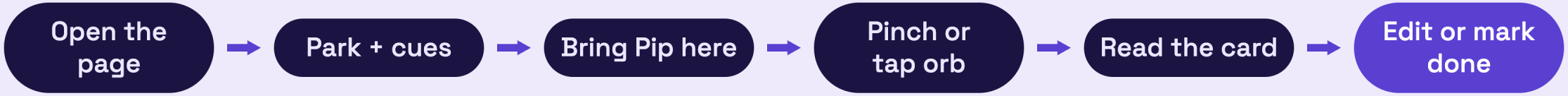
STAGE 5

Resume

Pinch or tap, read the card

> “This is what I need to remember when I come back.”

TASK FLOW

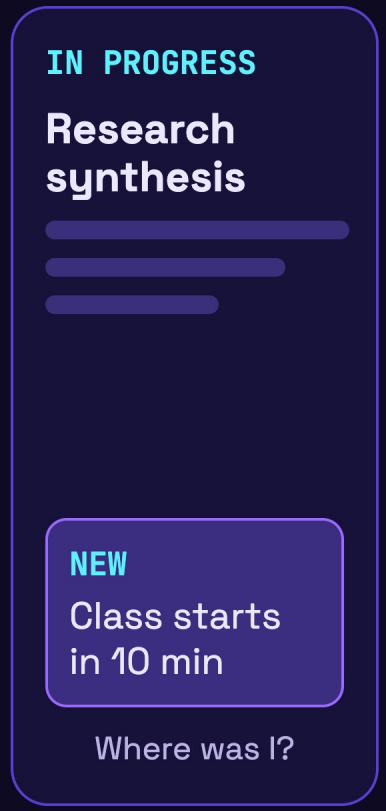


Quotes are the user’s thoughts at each stage, written for the journey map.

05 · STORYBOARD

Pause, park, bring Pip

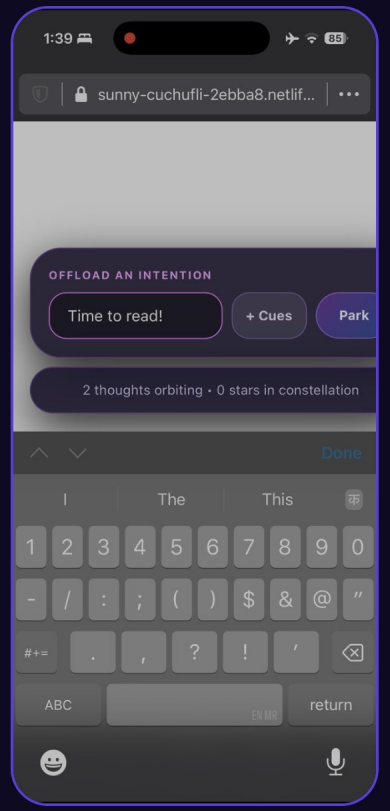
Frames 01–04 take one person from an interruption to Pip standing on their desk, with the thoughts orbiting.



01

Pause

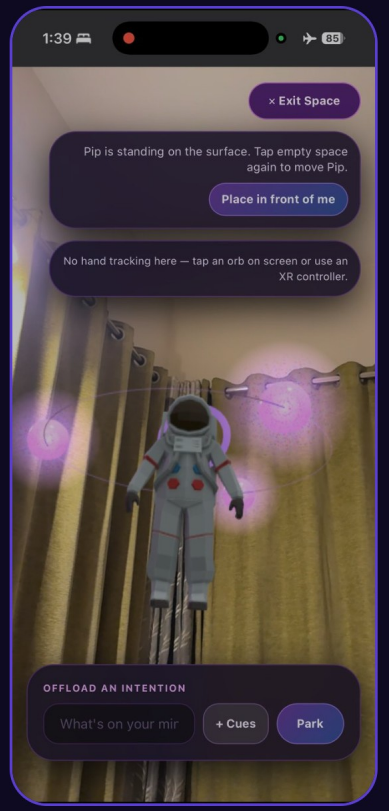
Interrupted mid-task. A plain to-do keeps only the title.



02

Park

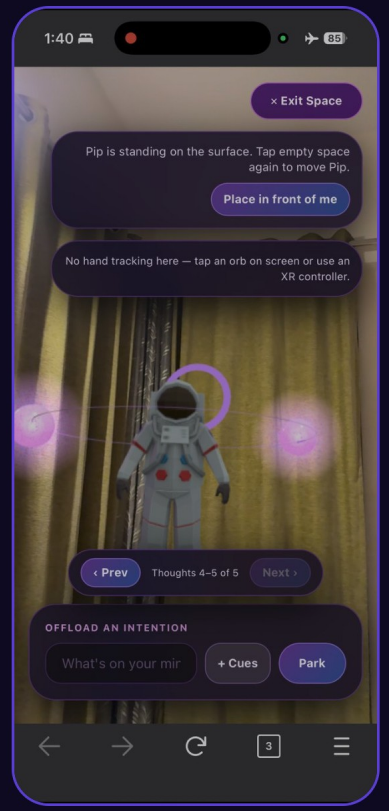
Type the thought and tap Park. Add cues if there is time.



03

Bring Pip

Pip stands on the aimed surface, or in front of you.



04

See the orbs

Each thought orbits Pip as an orb, three to a page.

Frame 01 is illustrated. Frames 02–04 are screens from the working prototype.

05 · STORYBOARD

Select, recall, resume

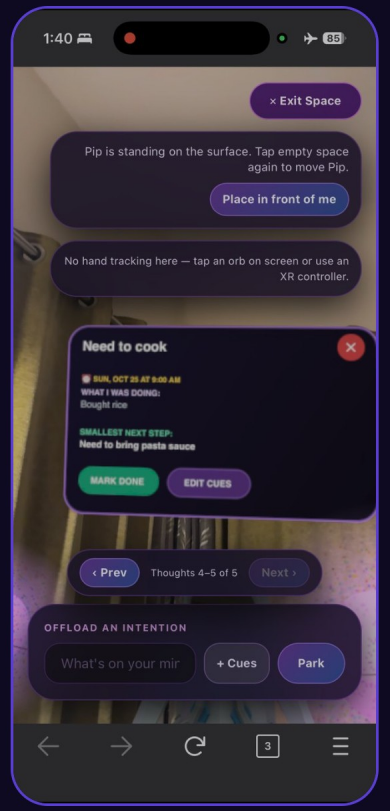
Frames 05–08 follow the return: pick an orb, read the thread, edit it, and close the card.



05

Select

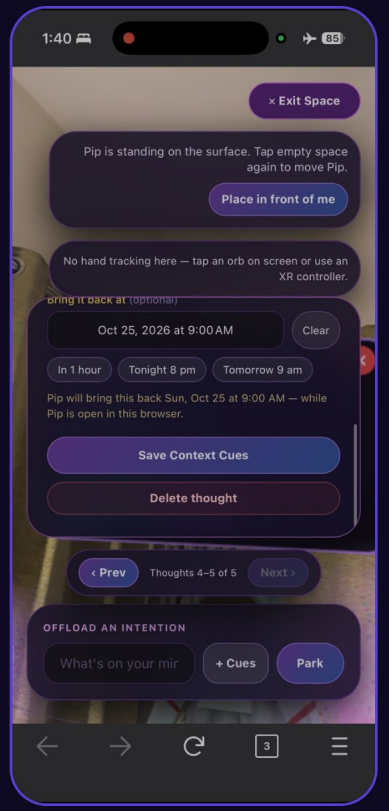
Pinch with hand tracking, or tap an orb. A ring marks it.



06

Recall

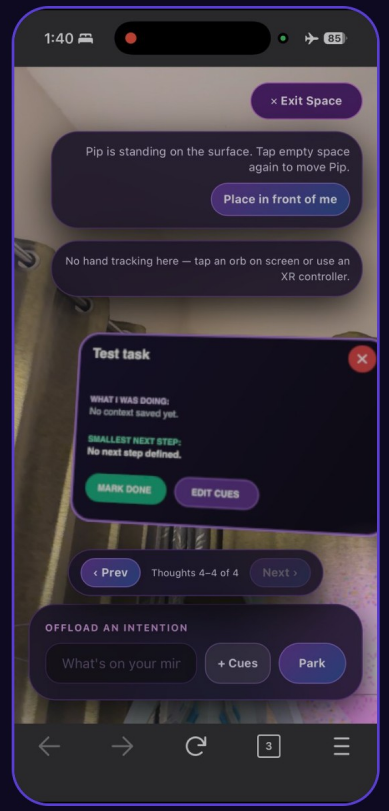
The card shows what you were doing and the smallest next step.



07

Edit cues

Add context and an optional bring-back time.



08

Close or finish

Close the card, or Mark Done to turn a thought into a star.

Frames 05–08 are screens from the working prototype.

06 · PROTOTYPE

A working WebAR prototype

sunny-cuchufli-2ebba8.netlify.app

WHAT IT DOES

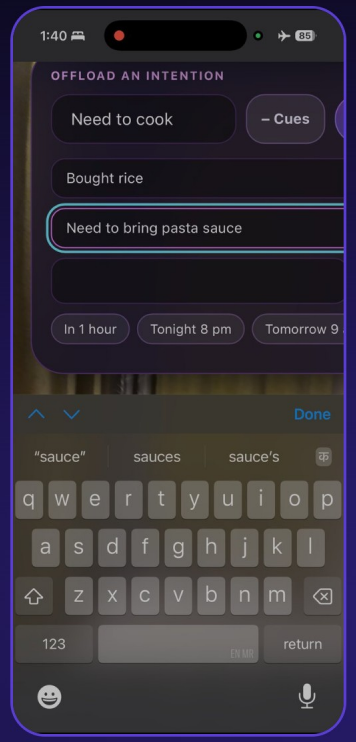
Parks each thought as an orb around a 3D astronaut. Optional cues and bring-back time. Mark Done turns a thought into a star.

INPUT

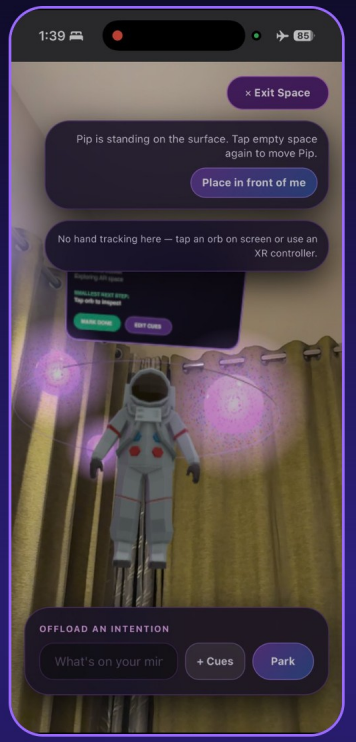
Pinch with XR hand tracking. Tap an orb on phones and in the 3D preview.

STATUS

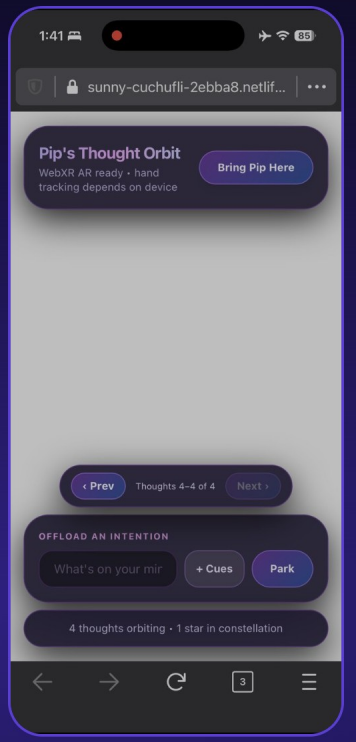
Runs in the browser and saves on the device. No backend or account. Bring-back times fire only while Pip is open.



Park with cues
Context, next step, optional time.



Pip in AR
Tap an orb to open its card.



Back on the page
Counters show thoughts and stars.

Screens captured on a phone from the live prototype.

07 · RATIONALE

Why AR, and how it stays safe

AR VALUE

A place, not a list line

Each thought gets a spot in the room you can look at. The orbs are the visual cue the experiment pointed to. Whether that beats a plain list is untested.

ACCESSIBILITY

Tap works everywhere

Hand tracking is optional: any orb opens with a tap, and Place in front of me needs no surface. Not yet checked with a screen reader.

SAFETY

Short and stationary

Pip stands on a surface in front of you, so nobody needs to walk while looking. Exit Space stays on screen. Not meant for walking or cooking.

PRIVACY

Stays on the device

No account and no backend; thoughts live in the browser's local storage. Orb titles show in the room, so keep them short and private.

These are design intentions, not tested outcomes.

08 · WHAT I WOULD TEST NEXT

Does the orbit beat a plain list?

Ask 3–5 design students to park the same three tasks in Pip and in a plain list, switch to another task for 10 minutes, then resume.

Capture
Time and taps to park a thought

Retrieval
Can they say what they were doing, and the next step

Input
Pinch versus tap: misses and retries

Comfort
Arm fatigue, glare, and how safe it feels

NOT MEASURED YET

How much effort capture really takes

Whether people choose to offload when busy

Whether the task actually gets done afterwards

So far: 1 interview, a 4-person experiment, no usability test.